ITA : 0501 COMPUTER VISION

CO : 2 - AT 3 - Class test

L B.S. RESHMA

L (192521342)

1. Digital Image Processing - Image Enhancement using Log and Gamma Transformation

A: Aim:

To brighten a dark museum photograph and reveal details in shadow regions

Method:

log transformation or power-law transformation

Steps:

1. Read the dark image and convert it into grayscale if required.
2. Normalize pixel values to the range 0 to 1.
3. Apply a log transformation or gamma transformation
4. Adjust the output intensity and display the enhanced image.

Formulas:

log transformation:

$$S = c \log(1 + r)$$

$$S = cr^{\gamma}$$

Result:

The dark photograph becomes brighter and exhibit details become more visible.

2. Histogram Equalization for fingerprint Enhancement:

A: Aim:

To improve fingerprint image and convert it to grayscale.

Method: Histogram Equalization

Steps:

1. Read the fingerprint image and convert it to grayscale
2. Calculate the histogram of the image
3. Compute the normalized histogram and cumulative distribution function.
4. Map each input intensity to a new intensity using CDF.
5. Display the enhanced fingerprint image.

Formula:

$$S_k = (L-1) \sum_{j=0}^{k} P(r_j)$$

Result: Fingerprint ridges and valleys become clearer, improving visual inspection.

3. Salt and pepper noise removal using Median filter:

Aim:

To remove salt and pepper while preserving edges.

Methods:

Median Filtering

Steps:

1. Read the noisy image
2. Select a 3x3 neighborhood around each pixel.
3. Arrange the neighborhood pixel values in ascending order.
4. Replace the centre pixel with median value
5. Repeat for all pixel and display filtered image.

~~Steps~~:

~~Rea~~

Formula:

$$g(x,y) = \text{median}\{f(s,t) : (s,t) \in S_{xy}\}$$

Result:

Salt and pepper noise is reduced while important object edges remain relatively sharp.

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4. Image Sharpening using Laplacian Filter:

Aim:

To enhance copper tracks and edges in blurred circuit board

Method:

Laplacian-based filter

Steps:

1. Read the circuit board image
2. Convert it to grayscale
3. Apply Laplacian operator to detect rapid intensity chan
4. Combine the laplacian image with the original image.
5. Display the sharped image.

Laplacian Mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Formula: $g(x,y) = f(x,y) + \nabla^2 f(x,y)$

Result:

Copper tracks and circuit board edg become sharper and easier to inspect.

5. Noise Reduction using Frequency Domain and Low pass filter.

A: Aim: To reduce high frequency graing noise in a drone image.

Method: Gaussian low-pass filtering

Steps:
1. Read the image and convert it to grayscale
2. Compute its two-dimensional Fourier transform
3. Shift the zero-frequency component
4. Apply a Gaussian low pass filter in frequency domain
5. Compute the inverse Fourier transform to obtain the smoothed image.

Formula: $H(u,v) = e^{-D^2(u,v)/(2D_0^2)}$

6. Blood Vessal Enhancement using high ~~butter~~ Pass filtering.

Aim: To enhance thin blood vessels and fine details in a retinal image

Method: High-boost filtering

Steps:
1. Read the retinal fundus image.
2. Convert it to grayscale
3. Apply a low-pass filter to obtain a blurred version
4. Subtract the blurred image from the original to

Obtain high frequency details.

5. Add the enhanced details back to the original image

Formula: $g(x,y) = Af(x,y) - f_L(x,y)$

Result: Thin blood vessels and other fine retinal details become more distinguishable from background.

7. Point and line Detection for bottle inspection:

Ans: Aim: To detect tiny air bubbles and hairline cracks on glass

method: Point detection and line detection masks

A) Point detection: $\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$

B) Line detection: Horizontal: $\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$; Vertical: $\begin{bmatrix} -1 & +2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$

Steps:

1. Convert the grayscale image
2. Apply point detection mask to identify isolated intensity changes and apply horizontal, vertical or diagonal line masks to identify thin cracks
3. Mark detected defects on the original image

Result: Isolated bubbles and thin cracks are highlighted.

8. Handwritting Extraction using Adaptive thresholding:

Aim: To separate dark handwriting with an even illumination

Method: Adaptive thresholding

Steps:

1. Read the handwritten answer image and convert to grayscale
2. Divide the image into small neighborhoods
3. Calculate a local threshold for each neighborhood
4. Classify pixels below the local threshold as ink and the remaining pixels as background.

6. Display the resulting binary image.

Formula: $g(x,y) = \begin{cases} 1, & f(x,y) < T(x,y) \\ 0, & f(x,y) \geq T(x,y) \end{cases}$

Result: Handwritten text is separated from the background more effectively.

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9. **Lake Extraction using Region Growing:**

Aim: To extract a lake from satellite image.

method: Seed-based region growing

Steps:
1. Read the satellite image and select a seed pixel inside lake.
2. Examine its neighboring pixels and add it whose intensities are sufficiently similar to region's representative
3. Repeat the process and Display the segmented lake region.

Formula: $|I(x,y) - M_R| < T$

Result: Pixels belonging to the lake are grouped into connected region, separating it from surrounding terrain.

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10. **Edge linking and Boundary Detection for mechanical gears**

Aim: To Obtain a complete gear outline from broken edge segments.

method: Hough transform and boundary detection.

Steps:
1. Convert the gear image to grayscale and apply noise reduction
2. Use an edge detector to identify edge pixels
3. Apply Hough transform to identify geometrically aligned edges
4. Extract the connected boundary and display the computed gear outline

Formula:

$$P = x\cos\theta + y\sin\theta$$

Result:

The mechanical Gear image's boundary is detected successfully.